Capital Structure and Gearing

Operating gearing

- **Operating gearing measures the *responsiveness* of operating earnings due to the presence of fixed costs in the firms cost structure given the change in revenue.**
- ***High fixed costs = high operating gearing = business risk.***
- **Operating gearing can be measured as:**
- **Operating gearing =** $\left(\frac{Fixed\ costs}{Total\ costs}\right)$***100 or** $\frac{\%\ change\ in\ EBIT}{\%\ change\ in\ revenue}$
- **Operating gearing is mainly influenced by the type of the sector in which a business is operating.**

Activity 1

- Consider the following details :

	K
Sales revenue	150,000
Cost of sales	(100,000)
Operating profit (EBIT)	50,000
Interest on debt	(30,000)
Earnings before tax	20,000

- The cost of sale is 75% variable.
- Required

Show the effect of a 10% change in revenue on the operating profit, on earning before tax, and on the interest cover. Comment on the results.

Solution to activity 1 (effects of fixed costs and Interest on earnings)

If sales *increases by 10%* :	K	% increase	
Sales revenue (150000*1.1)	165,000	10%	
Less Variable cost of sales (75000*1.1)	(82,500)	10%	
Contribution	82,500	10%	
Less Fixed cost (100000-75000)	(25,000)	0%	
Operating Profit (EBIT)	57,500	15%	5%
Interest on debt	(30,000)	0%	
Earnings before tax	27,500	37.5%	22.5%

Solution to activity 1 (effects of fixed costs and Interest on earnings)

If sales decrease *by 10%* :	K	% decrease	
Sales revenue	135,000	10%	
Less Variable cost of sales	(67,500)	10%	
Contribution	67,500	10%	
Less Fixed cost	(25,000)	0%	
Operating Profit	42,500	15%	5%
Interest on debt	(30,000)	0%	
Earnings before tax	12,500	37.5%	22.5%

Analysis of activity 1

- **Operating gearing =** $\frac{Fixed\ costs}{Total\ costs}$ **= =** $\frac{25{,}000}{100{,}000}$ **= <u>25%</u>**
- **Shows the amount of fixed costs in the firms cost structure.**
- **<u>Alternatively :</u>**
- **Operating gearing =** $\frac{\%\ change\ in\ EBIT}{\%\ Change\ in\ revenue}$ **= =** $\frac{15\%}{10\%}$ **= <u>1.50 times</u>**
- **The operating gearing measures how much a company's operating income changes in response to a change in sales.**
- **This ratio assists analysts in determining the impact of any change in sales on company earnings.**

Analysis of activity 1 cont.

- Initial Interest cover $= \frac{Profit\ before\ Interest\ and\ tax\ (PBIT)}{Interest\ (I)} = \frac{50{,}000}{30{,}000} = \underline{1.67\ times}$
- **Post 10% sales increase:**
- Revised Interest cover $= \frac{PBIT}{Interest} = \frac{57{,}500}{30{,}000} = \underline{1.92\ times}$ (15% increase)
- **Post 10% Decrease in sales:**
- Revised Interest cover $= \frac{PBIT}{Interest} = \frac{42{,}500}{30{,}000} = \underline{1.42\ times}$ (15% decrease).

- The presence of fixed cost in the firm's cost structure has a gearing effect on the interest cover ratio has indicated by a more than proportionate change in the cover compared to the change in sales

Financial gearing

- **Financial gearing (risk) is a measure of the extent to which *debt finance is used in the capital structure.***
- **Thus, an <u>all equity financed business is an ungeared firm</u>.**
- **High level of debt finance = high financial gearing.**
- **Financial gearing can be measured as:**
- **Debt to Equity ratio =** $\left(\frac{Long\ term\ debt + preference\ share\ capital}{Shareholders\ funds}\right)*100$
- **A business is highly geared if the above ratio exceeds 100%.**

Financial gearing

- **Debt ratio** $=\left(\frac{Total\ debt}{Total\ Assets}\right)*100$
- **This shows how much of the firms' total assets are financed using borrowed funds.**
- **A lower ratio is better in this case.**
- **Equity ratio** $=\left(\frac{Total\ equity}{Total\ Assets}\right)*100$
- **This shows how much of the total assets are financed using equity finance.**
- **A higher equity ratio is better.**

Financial gearing cont.

- **Debt to capital employed** $= \left(\frac{Long\ term\ debt + preference\ share\ capital}{Capital\ employed} \right) * 100$
- **Capital employed is the sum of debt and equity.**
- ***A firm is highly geared if the above ratio exceeds 50%***
- **The above ratios can be based on either market or book values.**
- **Interest cover (# of times)** $= \left(\frac{Profit\ before\ interest\ and\ tax}{Interest} \right)$
- **The ideal interest cover ratio is 3 times while a cover of 7 times is a safer cover.**

Managing and controlling gearing.

- **Debt management:**
  - **Reduce on the amount of debt by paying off certain loans**
  - **Avoid taking loans**
  - **Renegotiate the terms of the loans to reduce on the indebtedness.**
- **Increasing profits in order to push the share price up and subsequently the equity value of the firm.**
- **Reducing expenses in order to reduce the liabilities and subsequently lower gearing.**

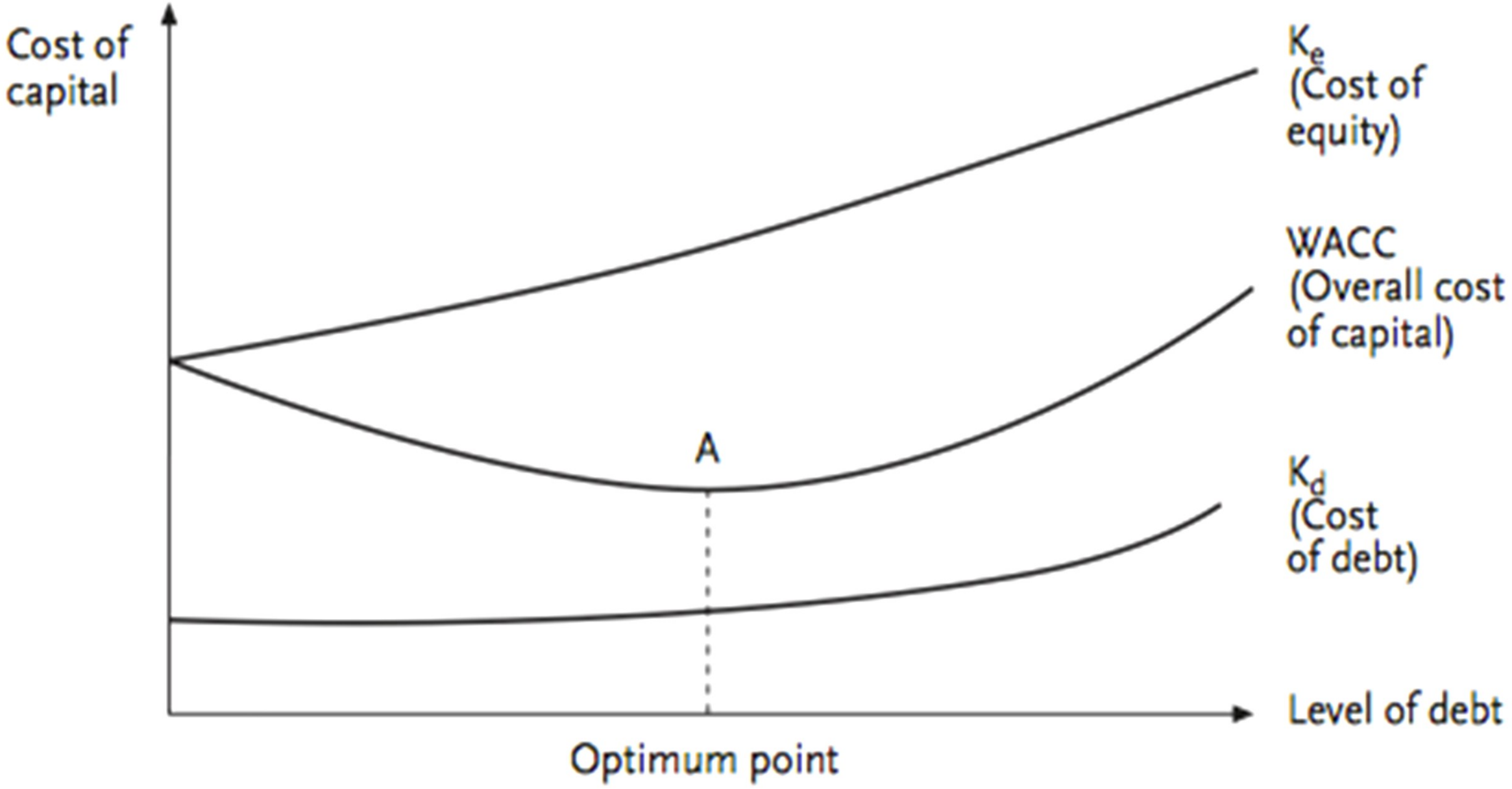
Capital structure theories

- ***Traditional or relevance theory*;**
- **Argues that the capital structure of a firm has an *influence on the company value*.**
- ***Increasing debt* at the *expense of equity* is expected to reduce WACC until the point at which *additional borrowing become expensive.***
- ***A lower WACC will increase the value of a project or a business.***
- ***Thus, the optimal capital structure is the mix of equity and debt at which WACC is lowest.***

Relevance theory or traditional view.

- **The point at which WACC is lowest is the level at which *company value is maximized.***
- **Therefore, if *<u>changing the mix of debt and equity will increase the firm value,</u>* managers will strive for that.**
- **Since the value of the firm is the sum of the equity value and the debt value, maximizing these values will maximize the firm.**
- **This can be achieved by minimizing the overall cost of capital which will give higher present values for all future cash flows.**

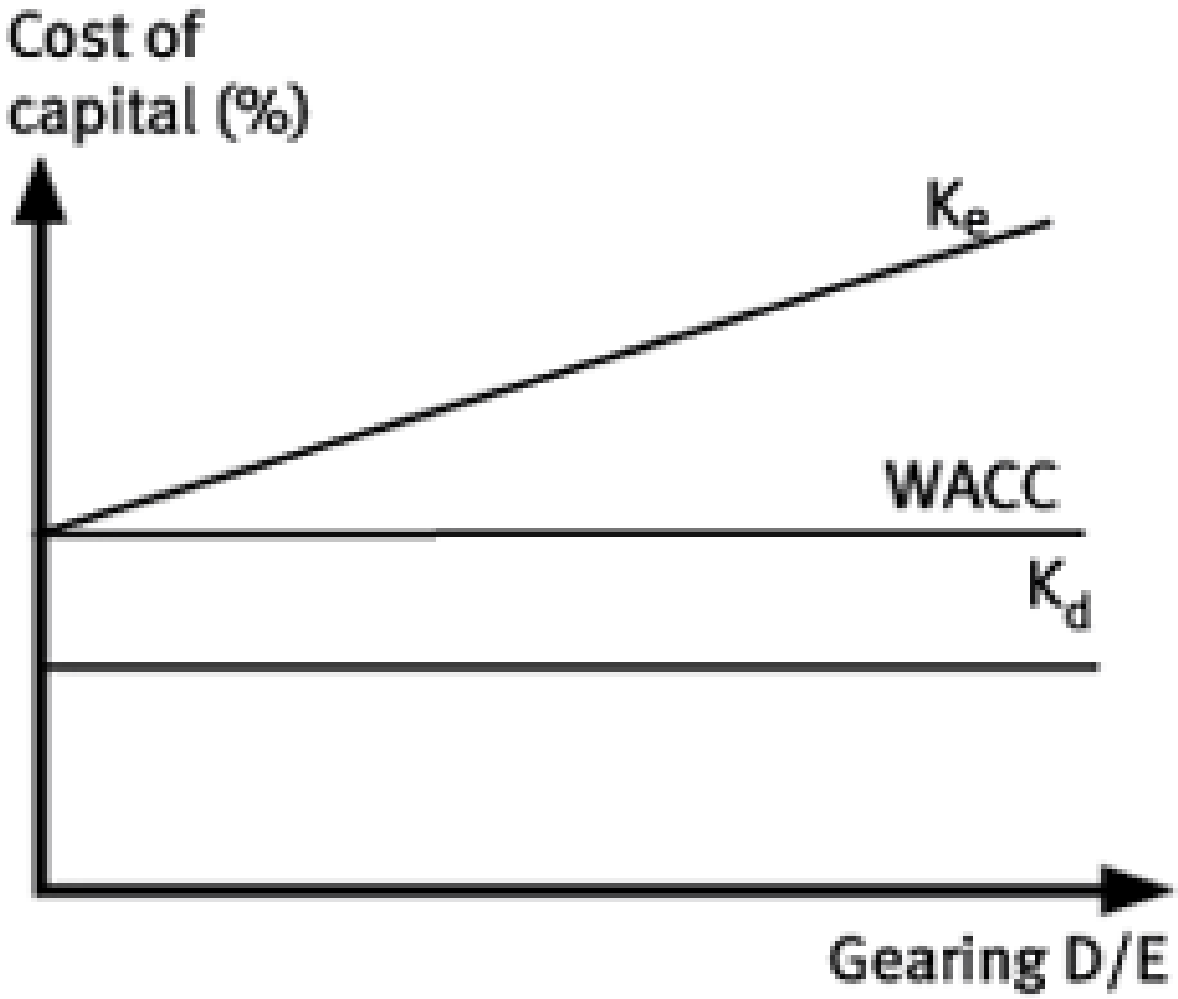
Traditional view.

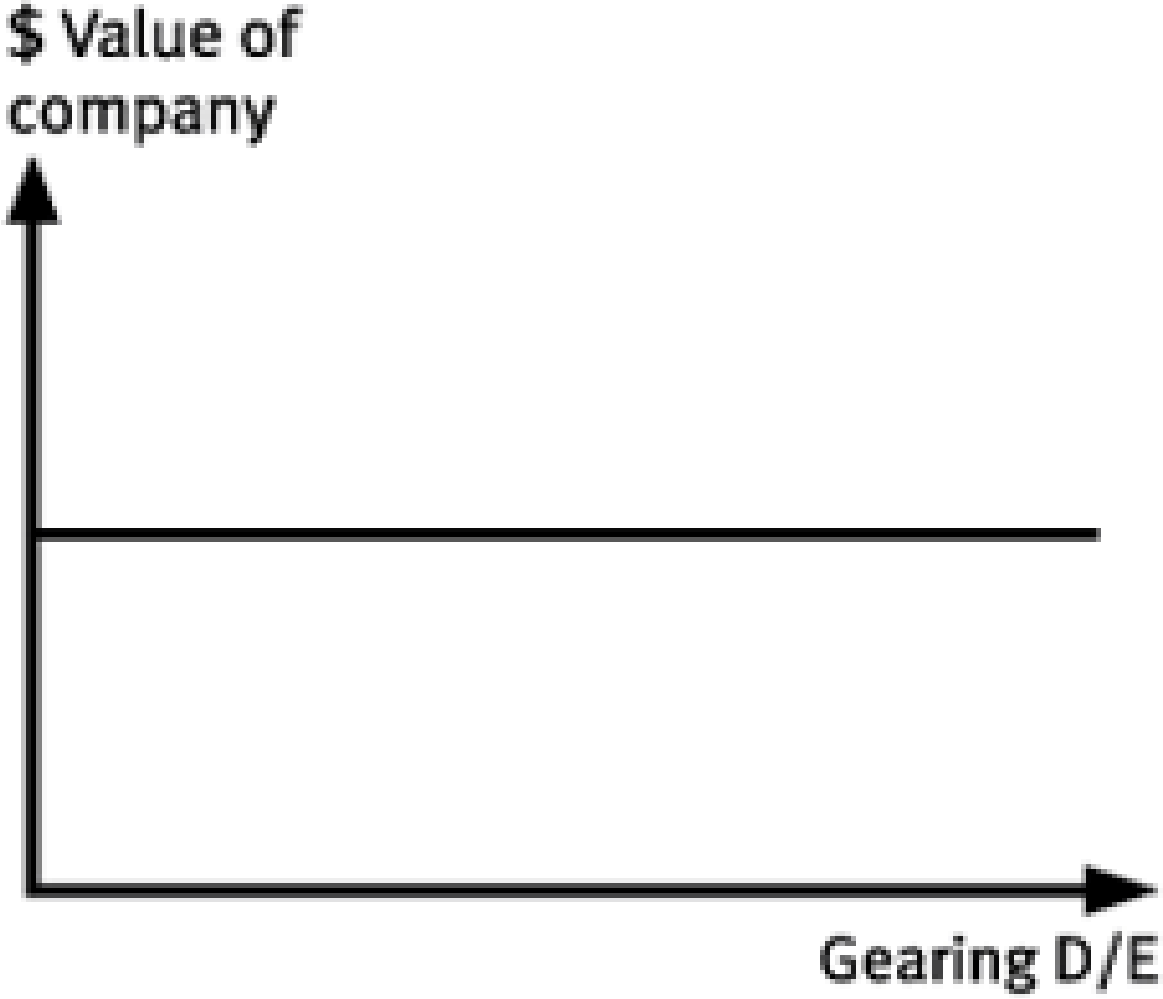


Irrelevance theory (M and M theories)

- ***Irrelevance theory (without tax);***
- **M and M argued that in a perfect capital market, the *capital structure is irrelevant.***
- **The *benefits of using cheap debt* is cancelled out by the increase in *cost of equity associated with financial risk*.**
- ***Thus, WACC remains unchanged and so is firm value.***
- ***Firm value can only be enhanced by investing in viable projects from the NPV perspective* and not through financing mix.**

Irrelevance theory (without tax).

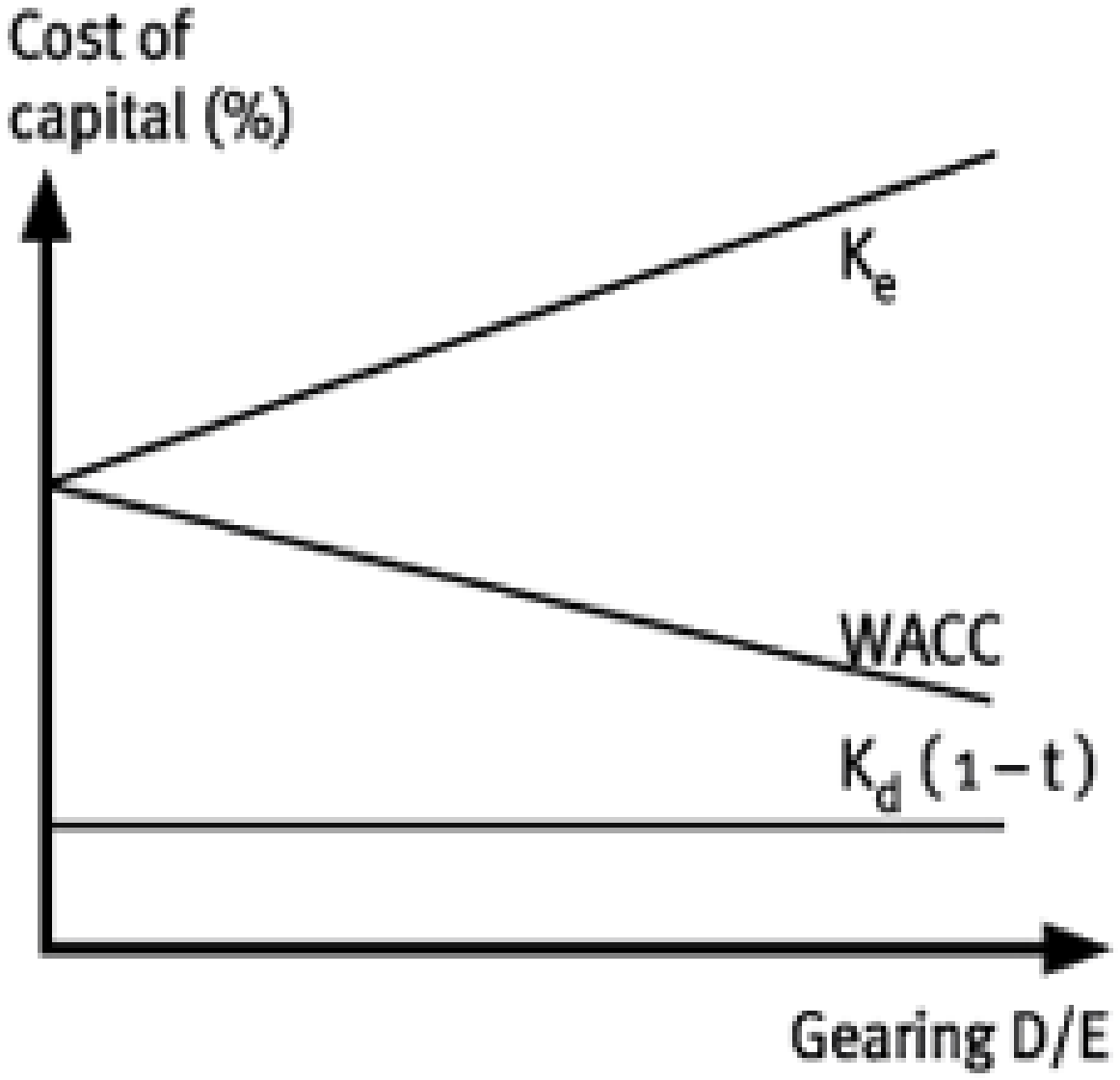


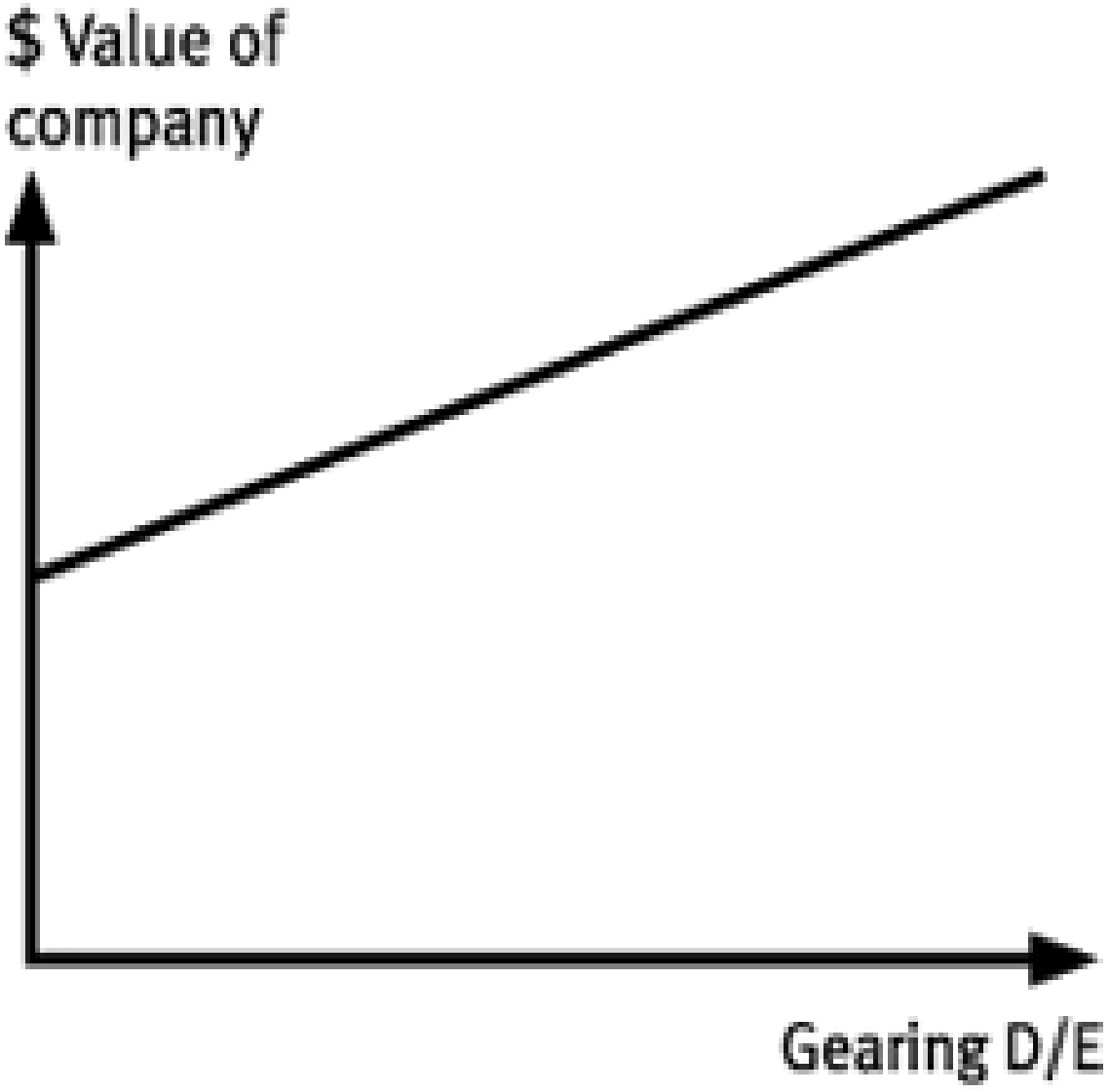


M and M with tax

- ***Irrelevance theory (with tax)*;**
- **Argued that *interest tax shield* will cause firms to use as much debt as possible.**
- ***WACC will keep on declining with increase in debt until the maximum level of debt is utilized.***
- **Company value will increase with increase in interest tax shield.**
- ***Value geared = Value ungeared + interest tax shield***

Irrelevance theory (with tax).





M and M with costs of financial distress

- ***Compromised view*; Argued that at *extreme gearing levels, financial distress cost become high thereby increasing WACC.***
- **<u>Thus the optimal capital structure is the mix of debt and equity just before financial distress sets in.</u>**
- **The above position is consistent with the traditional view.**

Pecking order theory

- This theory suggests that companies will prefer retained earnings to any other source of finance, and will opt for debt capital and equity as the last option.
- Business follow the order of least resistance in sourcing for long term financing

Pecking order theory cont.

Retained Earnings → Increasing risk and cost of capital

Debt finance → Increasing risk and cost of capital

Equity issuance → Increasing risk and cost of capital

Order of preference will be as follows.

- Retained
- Straight debt
- Convertible debt
- Preference shares
- Ordinary shares